

REPORT 08 - Scrum Plan

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1. OVERVIEW

1.1 Project Overview

Bank Associate Nation Keeper, Inc (B.A.N.K., Inc) has hired our company for building a mobile application that will help homeowners stay on top of their home maintenance. The mobile application includes features as follows:

- Monitors electricity usage,
- Monitor appliance upkeep,
- Monitor yard maintenance,
- Track periodic cleaning.

The application will provide a monthly checklist of home maintenance activities, allow the homeowner to check off items when they are complete, and produce a report at the end of the month.

Due to the Agile methodology's emphasis on working software (Beck et al. 2001), and the Scrum Management Framework practice of "incremental, iterative work cadences, known as sprints" (J. G., 2013), our software will be developed through several sprints, each concluding with deployment (James, 2012). Our project will consist of five 2-week iterations, resulting in a total time of approximately 10 weeks.

1.2 Project Specifics and Assumptions

- We are building a mobile application.
- Business requirements have been included in the contract.
- Pre-contract meetings took place with customer representatives.
- More meetings will be held throughout the development of the application, including a *Product Requirements Meeting*, following best Scrum practices and guidelines.

1.3 Implemented Software Development Methodology and Framework

1.3.1 Agile Software Development Methodology: Overview

In the development of this application, our company will implement the software development methodology known as Agile. This methodology was elaborated by *The Agile Alliance* (a group of 17 software practitioners) somewhere during 2001. This team documented in the *Manifesto for Agile Software Development* their following four main principles for Software Development:

- **Individuals and interactions** over processes and tools
- **Working software** over comprehensive documentation
- **Customer collaboration** over contract negotiation
- **Responding to change** over following a plan

1.3.2 Scrum Management Framework for Agile: Overview

In the development of this application, our company will implement the Agile Methodology through the Scrum Framework. This framework is succinctly explained using the following

diagram:

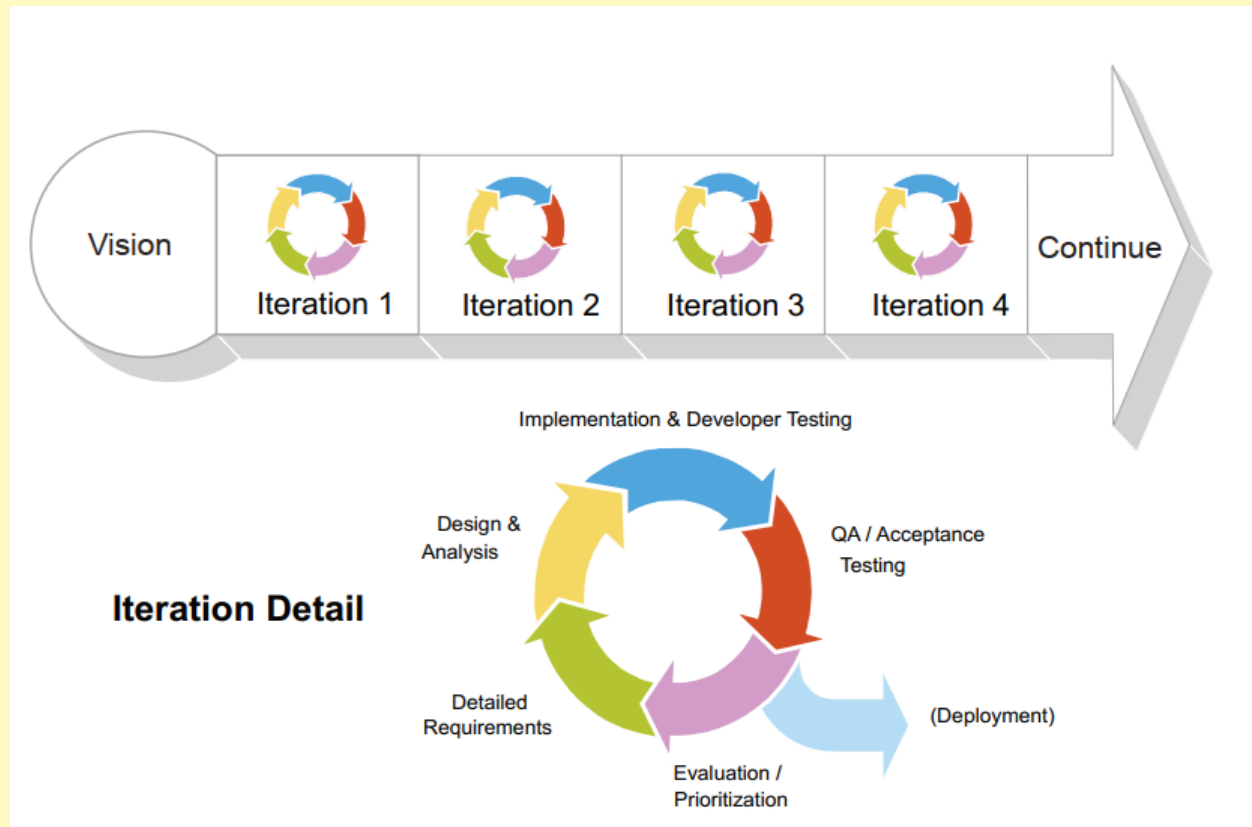


Figure 1 - Scrum blends all development activities into each iteration, adapting to discovered realities at fixed intervals (James, 2012).

This methodology is defined as follows, “Scrum is a lightweight framework that helps people, teams and organizations generate value through adaptive solutions for complex problems.” (Beck et al., 2002).

We will describe this framework in the following subsections (1.3.2.1 through 1.3.2.5). For additional description refer to Sutherland & Schwaber (2020).

1.3.2.1 Scrum: Theory (Pillars)

Events (1.3.2.4) and Artifacts (1.3.2.5) are possible thanks to the following pillars:

- **TRANSPARENCY** - The emergent process and work must be visible to those performing the work as well as those receiving the work.

- INSPECTION - Scrum artifacts and the progress toward agreed goals must be inspected frequently and diligently to detect potentially undesirable variances or problems.
- ADAPTATION - If any aspects of a process deviate outside acceptable limits or if the resulting product is unacceptable, the process being applied or the materials being produced must be adjusted (or adapt).

1.3.2.2 Scrum: Values

Successful use of Scrum depends on people becoming more proficient in living five values:

- COMMITMENT - The Scrum Team commits to achieving its goals and to supporting each other.
- FOCUS - Their primary focus is on the work of the Sprint to make the best possible progress toward these goals.
- OPENNESS - The Scrum Team and its stakeholders are open about the work and the challenges.
- RESPECT - Scrum Team members respect each other to be capable, independent people, and are respected as such by the people with whom they work.
- COURAGE - The Scrum Team members have the courage to do the right thing, to work on tough problems.

1.3.2.3 Scrum: Team

The fundamental unit of Scrum is a small team of people, referred to as a Scrum Team. The

Scrum Team consists of one ScrumMaster, one Product Owner, and a team of Developers.

Within a Scrum Team, there are no sub-teams or hierarchies. It is a cohesive unit of professionals focused on one objective at a time, the Product Goal.

- **DEVELOPERS** - Developers are the people in the Scrum Team that are committed to creating any aspect of a usable Increment each Sprint.
- **PRODUCT OWNER** - The Product Owner is accountable for maximizing the value of the product resulting from the work of the Scrum Team. How this is done may vary widely across organizations, Scrum Teams, and individuals.
- **Scrum MASTER** - The Scrum Master is accountable for establishing Scrum as defined in the Scrum Guide. They do this by helping everyone understand Scrum theory and practice, both within the Scrum Team and the organization.

1.3.2.4 Scrum: Events

The Sprint is a container for all other events. Each event in Scrum is a formal opportunity to inspect and adapt Scrum artifacts. These events are specifically designed to enable the transparency required. Failure to operate any events as prescribed results in lost opportunities to inspect and adapt. Events are used in Scrum to create regularity and to minimize the need for meetings not defined in Scrum.

- **THE SPRINT** - Sprints are the heartbeat of Scrum, where ideas are turned into value. They are fixed length events of one month or less to create consistency. A new Sprint starts immediately after the conclusion of the previous Sprint.
- **THE SPRINT PLANNING** - Sprint Planning initiates the Sprint by laying out the work to be performed for the Sprint. This resulting plan is created by the collaborative work of the entire Scrum Team.
- **DAILY Scrum** - The purpose of the Daily Scrum is to inspect progress toward the Sprint Goal and adapt the Sprint Backlog as necessary, adjusting the upcoming planned work.

- SPRINT REVIEW - The purpose of the Sprint Review is to inspect the outcome of the Sprint and determine future adaptations. The Scrum Team presents the results of their work to key stakeholders and progress toward the Product Goal is discussed.
- SPRINT RETROSPECTIVE - Each Sprint ends with a retrospective. At this meeting, the team reflects on its own process. They inspect their behavior and take action to adapt it for future Sprints (James, 2012).

1.3.2.5 Scrum: Artifacts

Scrum's artifacts represent work or value. They are designed to maximize transparency of key information. Thus, everyone inspecting them has the same basis for adaptation. Each artifact contains a commitment to ensure it provides information that enhances transparency and focus against which progress can be measured (Schwaber & Sutherland, 2020).

For the following description of Scrum artifacts, we will use the *Scrum reference card* by James, M. (2012).

- BACKLOG ARTIFACTS:
 - PRODUCT BACKLOG - Forced-ranked list of desired functionality, visible to all stakeholders and maintained by the Product Owner.
 - PRODUCT BACKLOG ITEM (PBI) - Specific functionality. Specifies the *what* more than the *how* of a customer-centric feature. Effort is estimated by the team, roughly 2-3 people in 2-3 days.
 - SPRINT BACKLOG - Consists of committed PBIs negotiated between the team and the Product Owner during the Sprint Planning Meeting.
 - SPRINT TASK - Specifies *how* to achieve the PBI's *what*. Requires one day or less of work.

- CHART ARTIFACTS:
 - SPRINT BURNDOWN CHART - It's a visual chart that indicates total remaining team task hours within one Sprint. It may not be used.
 - PRODUCT/RELEASE BURNDOWN CHART - Tracks the remaining Product Backlog effort from one Sprint to the next. Depicts historical trends to adjust forecasts.

1.3.2.6 Summary

The following items summarize the previous descriptions:

- Scrum is a management framework for incremental product development using one or more cross-functional, self-organizing teams of about seven people each.
- It provides a structure of ROLES, MEETINGS, RULES, and ARTIFACTS. Teams are responsible for creating and adapting their processes within this framework.
- Scrum uses fixed-length iterations, called Sprints, which are typically two weeks or 30 days long. Scrum teams attempt to build a potentially shippable (properly tested) product increment every iteration

These items (James, 2012) describe what we will integrate to the Scrum plan of our project.

Keywords: software development methodologies, Agile, management framework
Scrum

2. MEETINGS

2.1 Product Requirements Meeting

2.1.1 Who

- *ScrumMaster, Product Owner, Stakeholders*

2.1.2 Agenda

- *Welcome*
- *Discussion of project specifics*
- *Definition of software requirements*
- *Documentation of user stories*
- *Creation of Product Requirements Document*

2.1.3 Purpose

- *The purpose of the Product Requirements Meeting is to identify the specifics of the project to be reached over several iterations. Working with the stakeholders, project and software requirements will be identified and documented. User stories will also be discussed and documented. This meeting will result in the creation of the Product Requirements document, which will contain this information for reference over the course of the project (Radigan n.d). Though the Products Requirement Meeting is not required in James's Scrum Reference Card, we have chosen to implement a Products Requirement Document and Meeting for the mutual benefit of our stakeholders and company.*

2.1.4 Event

- *Held once at the beginning of the project.*

2.2 Sprint Planning Meeting

2.2.1 *Who*

- *All employees*

2.2.2 *Agenda*

- *Welcome*
- *Backlog refinement (as needed; only necessary if the Backlog Refinement Meeting was insufficient)*
- *Declaration of important Backlog Items*
- *Backlog Item selection*
- *Conversion of Backlog Items into Sprint Tasks.*

2.2.3 *Purpose*

- **The purpose of the Sprint Planning Meeting is to select which Product Backlog Items will be converted into products during the sprint. The Product Owner will decide which items have the highest priority, and the Development Team decides the amount of work they feel they can handle during the sprint. Items pulled from the Product Backlog will be placed in the Sprint Backlog (James 2012).**

2.2.4 *Event*

- *Held once at the beginning of the sprint.*

2.3 Daily Scrum

2.3.1 *Who*

- *Scrum Development Team, Product Owner (if needed)*

2.3.2 *Agenda*

- *Welcome*
- *Report of activities*

2.3.3 Purpose

- *As a rule, the Scrum Development Meeting must be short. In fact, Daily Scrum Meetings must be 15 minutes or less. Each team member will give a brief report of yesterday's activities, what they will do today, and what obstacles they are facing. The Sprint Task List and Sprint Burndown Chart may be updated at this meeting. The Product Owner may attend, but it is preferred that Daily Scrum only consists of the Development Team (James, 2012). As such, we have chosen to make the Product Owner's attendance optional, only if necessary.*

2.3.4 Event

- *Recurring, held every day in a designated, unchanging time and place.*

2.4 Sprint Review Meeting

2.4.1 Who

- *All employees, client representatives Rick and Rachel*

2.4.2 Agenda

- *Welcome*
- *Live demonstration of working product increment*
- *Declaration or rejection of commitments.*

2.4.3 Purpose

- **This meeting is a live demonstration of the completed product increment to the Product Owner, client, and all interested parties. After the**

demonstration, the Product Owner will review the commitments made at the Sprint Planning Meeting and declare them as finished or unfinished.

Incomplete items will be placed in the Product Backlog.

2.4.4 Event

- *Held once at the end of each sprint.*

2.5 Sprint Retrospective Meeting

2.5.1 Who

- *All employees.*

2.5.2 Agenda

- *Welcome*
- *Reflection of previous sprint*
- *Planning for future challenges*

2.5.3 Purpose

- **Every sprint will end with a Sprint Retrospective Meeting.** *The purpose of a retrospective meeting is to reflect on the previous iteration and create solutions for flaws and areas of improvement (Tripani, 2023). As such, The previous sprint's work will be discussed, and areas of concern will be noted. Plans will be created to adapt to challenges (James, 2012). This will be done through collection of team feedback, allowing all team members the opportunity to share opinions on the previous sprint.*

2.5.4 Event

- *This meeting will be held at the conclusion of each iteration.*

2.6 Backlog Refinement Meeting

2.6.1 *Who*

- *All employees.*

2.6.2 *Agenda*

- *Welcome*
- *Backlog Item effort estimation and prioritization*
- *Separation of large user stories into manageable tasks*

2.6.3 *Purpose*

- **As most Product Backlog Items are large, hard to understand, and unwieldy, the Backlog Refinement Meeting is held to prepare the backlog for the Sprint Planning Meeting. The Development Team will estimate the amount of effort needed to complete Backlog Items, and the Product Owner will prioritize them. Large items will be split into manageable tasks that are easier to understand (James 2012).**

2.6.4 *Event*

- *This meeting will be held at the conclusion of each iteration, after the Sprint Retrospective Meeting.*

3. DOCUMENTS

3.1 Product Backlog

3.1.1 Author

- Anyone, including the stakeholders, may create an item for the Product Backlog.

The log is maintained by the Product Owner.

3.1.2 Intended Audience

- All employees and stakeholders.

3.1.3 Purpose

- **The Product Backlog is a collection of Product Backlog Items. In the backlog, items are listed in order of priority and ROI (James 2012).** The Product Backlog allows for simple, easily readable system design and a more loosely defined documentation strategy.

3.1.4 Deadline

- *This document will be created on the first day of the project at the first Sprint Planning Meeting. It will be continuously maintained by the Product Owner and refactored at the Backlog Refinement Meeting.*

3.2 Product Backlog Items

3.2.1 Author

- Anyone, including the stakeholders, may create a Product Backlog Item.

3.2.2 Intended Audience

- All employees and stakeholders, particularly the Development Team.

3.2.3 Purpose

- **Product Backlog Items (PBI) are often written like User Stories. They define a task to be completed and the necessary acceptance criteria. The Development Team will estimate how much time and manpower is required to complete each item (James 2012).** These items are similar to User Stories as detailed in the XP methodology. In Salimi's words, "*Product Backlog Items ensure that the customer is getting the right product to meet their needs*" (Salimi, n.d.)

3.2.4 Deadline

- *These documents will first be created on the first day of the project at the first Sprint Planning Meeting. After this, they will be re-evaluated, moved through the backlog, and even created daily.*

3.3 Sprint Backlog

3.3.1 Author

- PBIs found in the Sprint Backlog will be negotiated by the Product Owner and Development Team.

3.3.2 Intended Audience

- All employees and stakeholders, particularly the Development Team.

3.3.3 Purpose

- **Not to be confused with the Product Backlog, the Sprint Backlog is a collection of Product Backlog Items to be completed during the current Sprint. It details the status and requirements of tasks to be completed and finished tasks. It is visible to the team and discussed during Daily Scrum**

Meetings (James 2012). This artifact is similar to the Product Backlog, but refers more specifically to *only* PBIs to be worked on during the current sprint.

3.3.4 Deadline

- *This document will be created at the first Sprint Planning Meeting. It will be re-evaluated at each successive Sprint Planning Meeting.*

3.4 Sprint Task

3.4.1 Author

- Development Team

3.4.2 Intended Audience

- All employees and stakeholders, particularly the Development Team.

3.4.3 Purpose

- If PBIs are the *what* of a problem, Sprint Tasks are the *how*. Each Sprint Task is a broken down PBI requiring one day or less of work. Remaining effort will be re-estimated daily. The Development Team is expected to collaborate towards completion of each Task (James 2012).

3.4.4 Deadline

- *These documents will first be created on the first day of the project at the first Sprint Planning Meeting. After this, they will be re-evaluated, moved through the backlog, and even created daily.*

3.5 Sprint Burndown Chart

3.5.1 Author

- Development Team and ScrumMaster

3.5.2 Intended Audience

- All employees and stakeholders, particularly the Development Team.

3.5.3 Purpose

- **The Sprint Burndown Chart shows the remaining team task hours for each sprint. Because this information is ever-changing, it will be re-evaluated daily. It is a visual guide showing the amount of man-hours left in each iteration (James 2012). *Use of the Sprint Burndown Chart is not necessarily required in Scrum, however, we believe that the added detail the chart provides will be useful for our team.***

3.5.4 Deadline

- *This document will first be created on the first day of the sprint at the first Sprint Planning Meeting. It will then be re-evaluated daily.*

3.6 Product Burndown Chart

3.6.1 Author

- Product Owner

3.6.2 Intended Audience

- All employees and stakeholders

3.6.3 Purpose

- **Not to be confused with the Sprint Burndown Chart, the Product Burndown Chart shows the remaining team task hours and backlog effort over the course of each sprint. It will also act as a forecasting tool, displaying predictions for man-hours in future sprints. (James 2012). *Use of the Project Burndown Chart is especially beneficial to stakeholders and the Product Owner,***

as it visually details and predicts the amount of manpower needed to finish the project.

3.6.4 Deadline

- *This document will first be created on the first day of the sprint at the first Sprint Planning Meeting. It will then be re-evaluated after each sprint in the Sprint Retrospective Meeting.*

3.7 Products Requirement Document

3.7.1 Author

- *Product Owner, ScrumMaster, Stakeholders*

3.7.2 Intended Audience

- *All employees and stakeholders*

3.7.3 Purpose

- *A Product Requirements Document (PRD) will be created before the project is started. In Radigan's words, "It outlines the product's purpose, its features, functionalities, and behavior". A PRD will determine the specifics of the project, assumptions, and business objectives. It will also include user stories, which will be beneficial in the creation of Sprint Tasks and PBIs (Radigan, n.d). Though the Products Requirement Document is not required in James's Scrum Reference Card, we have chosen to implement a Products Requirement Document for the mutual benefit of our stakeholders and company.*

3.7.4 Deadline

- *This document will be created at the Product Requirements Meeting.*

3.8 Test Plan

3.8.1 Author

- *Development Team*

3.8.2 Intended Audience

- *All employees and stakeholders*

3.8.3 Purpose

- *The Scrum methodology, as detailed in James's 2012 "Scrum Reference Card", does not require a specific testing document. However, for the sake of organization, all tests created for this project will be organized into a test plan document. This document will be updated daily as new tests are run, completed, and created. This document will serve as a guide for all tests created in each iteration.*

3.8.4 Deadline

- *This document will be created at the beginning of each iteration. It will also be updated daily as tests are run and new tests are created.*

4. ROLES

4.1 Project Owner (1 person)

4.1.1 Who

- Oliver

4.1.2 Qualifications

- Significant experience in development, testing, and especially management. The Product Owner must be capable of communicating with clients and administrating the software development process.

4.1.3 Responsibilities

- **The product owner is responsible for maximizing ROI and product vision. Their duties include overseeing the Product Backlog, approving or rejecting increments or software, communicating with stakeholders, and leading the team (James 2012).** *They should also assist in drafting documentation and designing the finished product. The Project Manager will preside over the entire development process and aim the project towards completion (Ovcharenko 2024).*

4.2 ScrumMaster (1 person)

4.2.1 Who

- Abe

4.2.2 Qualifications

- Significant experience in development, testing, and especially management. The ScrumMaster must be capable of leading a team, overseeing relationships between team members, and managing artifacts.

4.2.3 Responsibilities

- **The ScrumMaster will facilitate the Scrum process. They will promote ideal development practices, lead the team to success, study product data to create predictions and forecasts, and keep artifacts visible. However, they must not hold any authority over the team - they are leaders, but not authority figures (James 2012). *Abe was chosen as the ScrumMaster due to his vast experience. Employees see him as a leader and project veteran, making him perfect for the ScrumMaster role.***

4.3 Scrum Development Team 1 (6 people)

4.3.1 Who

- Britney, Emily, Grace, Ingrid, Larry, Ursula

4.3.2 Qualifications

- College degree or working experience in software engineering, computer science, design, or a related field.

4.3.3 Responsibilities

- **The Scrum Development Team is cross-functional. It does not consist merely of developers, but also analysts, testers, and even designers. The team must be self-managing and willing to negotiate with the Product Owner. Additionally, team members must be collaborative and spend their full time in cooperation with one another. The team is responsible for much of the analysis, development, design, and testing of the project. It is recommended that Development Teams consist of around seven members, no more than nine (James 2012). *In light of this, we have split our employees into two separate Scrum Development Teams. This is Team One of two.***

4.4. Scrum Development Team 2 (7 people)

4.3.1 Who

- Claire, Doug, Frank, Holly, Jack, Keith, Xavier

4.3.2 Qualifications

- College degree or working experience in software engineering, computer science, design, or a related field.

4.3.3 Responsibilities

- **The Scrum Development Team is cross-functional. It does not consist merely of developers, but also analysts, testers, and even designers. The team must be self-managing and willing to negotiate with the Product Owner.**

Additionally, team members must be collaborative and spend their full time in cooperation with one another. The team is responsible for much of the analysis, development, design, and testing of the project. It is recommended that Development Teams consist of around seven members, no more than nine (James 2012). *In light of this, we have split our employees into two separate Scrum Development Teams. This is Team Two of two.*

5. CHECKPOINTS

5.1 *Product Requirements Completed*

5.1.1 Effort Estimation

- *1 week after the project starts.*

5.1.2 Exit Criteria

- *Project and software requirements have been identified and documented through collaboration with stakeholders. A Project Requirements Document has been created and to-be implemented user stories for this project have been defined. This checkpoint will only be completed upon creation of the Project Requirements Document. Though the Products Requirement Meeting is not required in James's Scrum Reference Card, we have chosen to implement a Products Requirement Document and Meeting for the mutual benefit of our stakeholders and company.*

5.2 Sprint Planning Completed

5.2.1 Effort Estimation

- *Recurring, will be completed on the first day of each sprint.*

5.2.2 Exit Criteria

- **A plan for the subsequent sprint has been completed including the Sprint Backlog. PBIs from the Product Backlog have been converted into Sprint Tasks, and the Development Team has negotiated the amount of work they feel they can handle during the sprint. This checkpoint can only be completed after the Sprint Planning Meeting.**

5.3 Implementation and Developer Testing Completed

5.3.1 Effort Estimation

- *Recurring - but at most, this will take 2 weeks per iteration.*

5.3.2 Exit Criteria

- **All Sprint Tasks as defined in the Sprint Plan have been completed. In other words, this iteration's development work is finished. Additionally, unit and developer tests have been completed.**

5.4 QA and Acceptance Testing Completed

5.4.1 Effort Estimation

- *This checkpoint will be progressing during the development checkpoint. As such, it will end at the same time as the development checkpoint - 2 weeks at maximum.*

5.4.2 Exit Criteria

- *All defined test cases, as defined in the test plan document, are successful and free from errors.* **The code is clean, readable, and deployable. It has been thoroughly debugged and rigorously tested. Additionally, the code must fulfill all safety and security requirements.**

5.5 Iteration Deployment

5.5.1 Effort Estimation

- *At maximum, 2 weeks after iteration starts.*

5.5.2 Exit Criteria

- **A Sprint Review Meeting has been held with stakeholders, including a live demonstration of the working product increment to the Product Owner, stakeholders, and all interested parties. Upon declaration of finished commitments from the Product Owner and with all development and testing**

complete, the software for this iteration is released. This checkpoint will only be fulfilled upon successful release of this iteration's software.

5.6 Retrospective Completed

5.6.1 Effort Estimation

- *This checkpoint will be reached at the conclusion of the iteration - at maximum, 2 weeks from iteration start.*

5.6.2 Exit Criteria

- The retrospective meeting has been completed, and all team members have reviewed the iteration's successes and failures. Additionally, the Product Backlog has been refined and reprioritized through the Backlog Refinement Meeting, and the Product Burndown Chart has been updated by the Product Owner. This checkpoint will only be fulfilled upon completion of the Sprint Retrospective and Backlog Refinement Meetings.

5.7. Timeline

Event	Date
<i>Product Requirements Completed</i> <u>Deliverable</u> : <i>Product Requirements Document</i>	1 week after the project starts.
Sprint Planning Complete <u>Deliverable</u> : Sprint Backlog	1 day after iteration starts.
Development and Testing Completed	At maximum, 2 weeks after iteration starts.
Iteration Deployed <u>Deliverable</u> : One Iteration's Completed, Working Software	At maximum, 2 weeks after iteration starts.

Iteration Retrospective Complete <u>Deliverable: Reprioritized Product Backlog</u> <u>and Product Burndown Chart</u>	2 weeks after iteration starts.
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6. ANALYSIS

In the following section, we will list the strengths and weaknesses of the Scrum Management Framework, following the analysis made by Vasiliauskas (2024). Then we will address a final summary and reflect on our implementation of the methodology.

6.1 Strengths/Advantages

1. **Adaptability and flexibility** - Scrum is itself a flexible and adaptable practice and is also suitable for a wide variety of environments and situations that require a flexible approach (for example, when it can be difficult to identify clear requirements).
2. **Creativity and innovation** - Scrum emphasizes creativity and new ideas. Because Scrum teams work closely together and analyze ideas from all members, this encourages new and innovative solutions.
3. **Time to market** - Scrum often results in faster delivery than other approaches, speeding time to market for new products or features.
4. **Lower costs** - The Scrum approach can be cheaper for an organization, as it usually requires less documentation and control than other project management approaches.
5. **Increased transparency often leads to higher quality work** - Scrum methodology ensures that team members are focused and productive, partly by creating transparency amongst teams, as well as between teams, internal stakeholders, and clients. All changes to any part of a project are visible to every member, no matter how small. This builds trust between all the parties. At the same time, because all team members take full responsibility and ownership of their work, this often creates a productive environment and high-quality end results.

6. **Continuous feedback improves customer satisfaction** - With daily standup meetings ensuring continuous feedback and constant adjustments to the product based on this feedback, products and solutions usually end up being more popular with clients and customers.
7. **Increases motivation and team satisfaction** - Because Scrum encourages personal responsibility from every team member, they tend to feel more ownership of and pride in the project, leading to greater motivation and satisfaction.

6.2 Weaknesses/Disadvantages

1. **You need experienced team members** - Because Scrum involves extended periods of intense work, all team members need to be able to successfully perform their own tasks. This generally requires enough experience to be able to do this. Everyone on the team needs to execute and provide quality feedback on the process and results. If you don't have this experience in your organization, then we move to disadvantage number two.
2. **It requires heavy training** - Without experience, your team needs training. This is time-consuming, as everyone on the team needs to understand the benefits, the artifacts, the thought process, and the practice. If you're starting from scratch, expect this to take months before your team is fully up to speed.
3. **It only really works with small teams** - Scrum usually works best with teams of no more than nine or ten. This can help make teamwork easier, larger organizations may be unused to splitting the workforce into small, autonomous teams, and may struggle to do so effectively, slowing their effective adoption of Scrum.
4. **It can be difficult to scale** - Related to the challenge above, implementing Scrum on a bigger scale usually requires extensive training and coordination. Scrum can be used on

larger projects with larger teams, but rarely is, due to this being so complex to implement and manage successfully.

5. **It doesn't necessarily help to hit a project's overall deadline** - Adopting Scrum means breaking larger projects into smaller, bite-sized pieces that are easier and quicker to implement. This in turn means multiple smaller deadlines. However, this will not necessarily lead to the project hitting its overall deadline. Project managers and stakeholders need to keep an eye on the larger project to ensure that it doesn't fall too far behind schedule.
6. **It can be hard to integrate with a classic project management approach** - The Scrum methodology is often poorly suited to projects that need predictability and a well-defined plan, for which a waterfall approach may work better. Scrum is great for flexibility and short-term planning, but not so much for long-term planning.
7. **It may require major transformation within an organization** - Related to the previous disadvantage, adopting the Scrum framework can mean that the organization needs to transform to accommodate the practice. Some parts of the process may require different departments to collaborate differently leading to significant organizational change.

6.3 Summary & Reflection

Overall, we think that this methodology is suitable to our needs. We count on a balanced team of experienced members and newcomers to the industry as well, all of them willing to learn and adapt to some of the challenges that this methodology may propose. Also, we can consider that this methodology is highly adaptable for teams of our size, so to some extent we have an ideal scenario for utilizing Scrum.

Additionally, our team will rely on continuous customer feedback from B.A.N.K.

representatives, Rick and Rachel. The Scrum methodology requires open communication and collaboration with stakeholders, which our team can easily facilitate.

The Scrum methodology makes efficient use of our resources due to its focus on self-organized team collaboration (James 2012). Working as a team, we will be able to quickly and accurately create a product that fulfills all stakeholder requirements. This keeps the Scrum plan low risk in our scenario, as most possible issues listed by James derive from poor team communication and collaboration. A dedicated full time ScrumMaster and culture of teamwork and collaboration will negate these risks (James 2012).

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